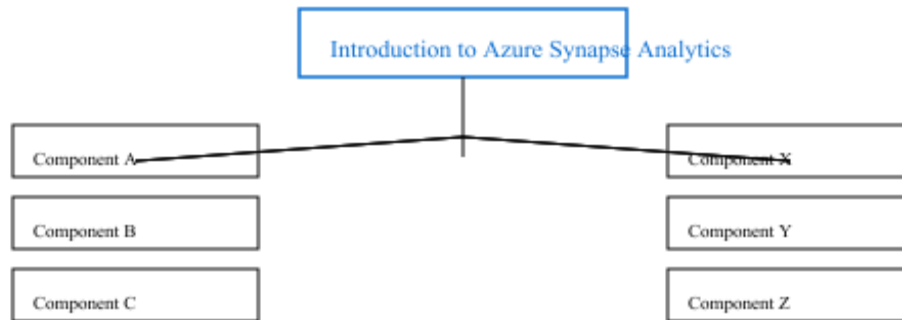


# Chapter 1: Introduction to Azure Synapse Analytics

This chapter covers: What is Synapse?, Why use Synapse?, Personas, Core Concepts, Quickstart Lab. It is written in a hybrid style — professional explanations with easy analogies and teaching tips.



## Section 1: What is Synapse?

Detailed explanation of What is Synapse?. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the What is Synapse? lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## Section 2: Why use Synapse?

Detailed explanation of Why use Synapse?. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Why use Synapse? lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## Section 3: Personas

Detailed explanation of Personas. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Personas lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## Section 4: Core Concepts

Detailed explanation of Core Concepts. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Core Concepts lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Section 5: Quickstart Lab**

Detailed explanation of Quickstart Lab. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

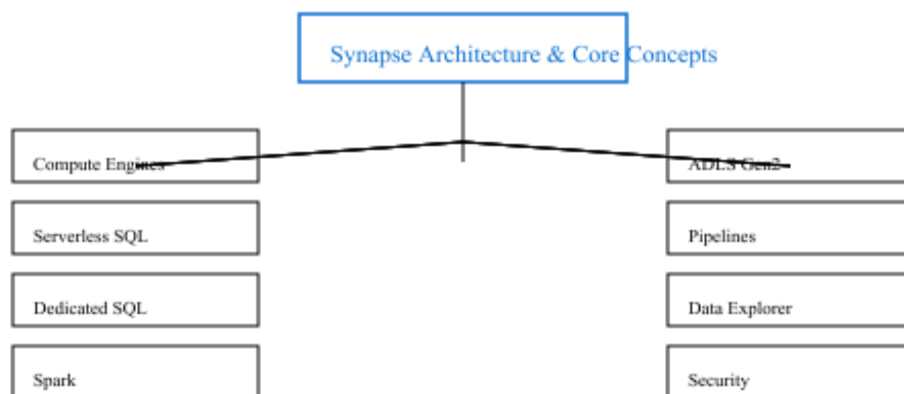
Lab: Perform the Quickstart Lab lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Interview Prep & Quick Notes**

Key points to remember for interviews: explain tradeoffs, give Azure services mapping, and discuss performance implications.

## Chapter 2: Synapse Architecture & Core Concepts

This chapter covers: Compute vs Storage, MPP internals, Lakehouse model, Data formats, Performance principles. It is written in a hybrid style — professional explanations with easy analogies and teaching tips.



### Section 1: Compute vs Storage

Detailed explanation of Compute vs Storage. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Compute vs Storage lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 2: MPP internals

Detailed explanation of MPP internals. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the MPP internals lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 3: Lakehouse model

Detailed explanation of Lakehouse model. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Lakehouse model lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 4: Data formats

Detailed explanation of Data formats. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Data formats lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Section 5: Performance principles**

Detailed explanation of Performance principles. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

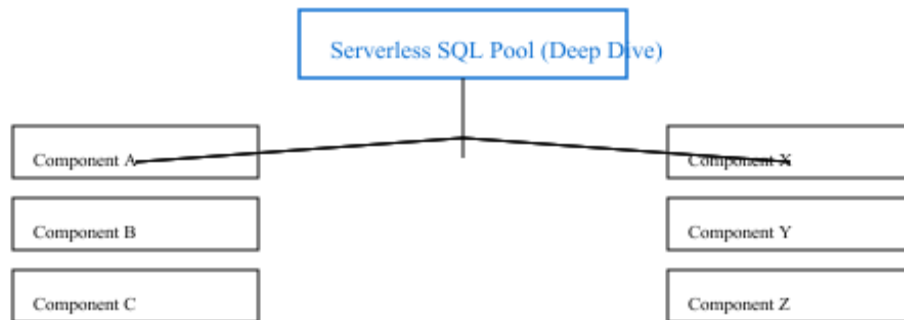
Lab: Perform the Performance principles lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Interview Prep & Quick Notes**

Key points to remember for interviews: explain tradeoffs, give Azure services mapping, and discuss performance implications.

## Chapter 3: Serverless SQL Pool (Deep Dive)

This chapter covers: OPENROWSET, External Tables, Partition Pruning, Cost control, Advanced queries & labs. It is written in a hybrid style — professional explanations with easy analogies and teaching tips.



### Section 1: OPENROWSET

Detailed explanation of OPENROWSET. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

#### Code Example:

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https://path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

Lab: Perform the OPENROWSET lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 2: External Tables

Detailed explanation of External Tables. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

#### Code Example:

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https://path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

Lab: Perform the External Tables lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 3: Partition Pruning

Detailed explanation of Partition Pruning. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

**Code Example:**

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https:///path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

Lab: Perform the Partition Pruning lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## Section 4: Cost control

Detailed explanation of Cost control. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

**Code Example:**

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https:///path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

Lab: Perform the Cost control lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## Section 5: Advanced queries & labs

Detailed explanation of Advanced queries & labs. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

**Code Example:**

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https:///path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

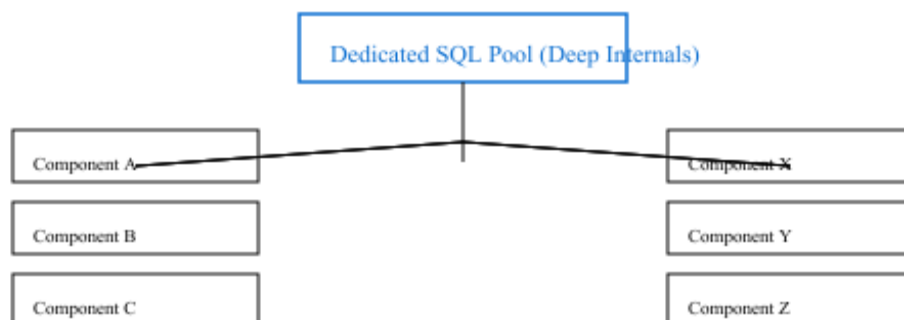
Lab: Perform the Advanced queries & labs lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Interview Prep & Quick Notes**

Key points to remember for interviews: explain tradeoffs, give Azure services mapping, and discuss performance implications.

## Chapter 4: Dedicated SQL Pool (Deep Internals)

This chapter covers: Distributions, Columnstore Index, CTAS, PolyBase & COPY INTO, Resource classes. It is written in a hybrid style — professional explanations with easy analogies and teaching tips.



### Section 1: Distributions

Detailed explanation of Distributions. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

#### Code Example:

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https://path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

Lab: Perform the Distributions lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 2: Columnstore Index

Detailed explanation of Columnstore Index. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

#### Code Example:

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https://path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

Lab: Perform the Columnstore Index lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 3: CTAS

Detailed explanation of CTAS. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

**Code Example:**

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https:///path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

Lab: Perform the CTAS lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## Section 4: PolyBase & COPY INTO

Detailed explanation of PolyBase & COPY INTO. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

**Code Example:**

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https:///path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

Lab: Perform the PolyBase & COPY INTO lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## Section 5: Resource classes

Detailed explanation of Resource classes. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

**Code Example:**

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https:///path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

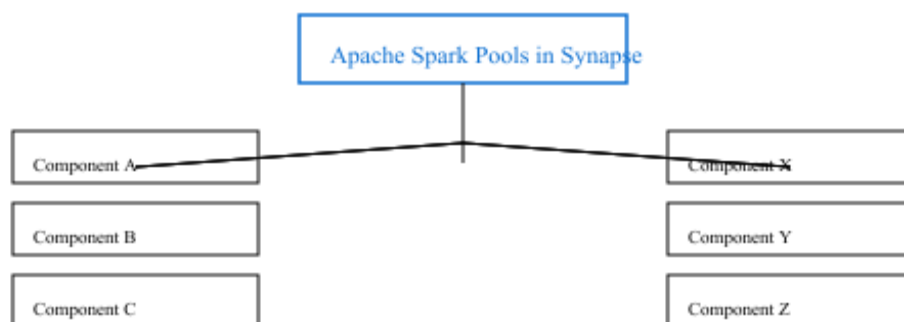
Lab: Perform the Resource classes lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Interview Prep & Quick Notes**

Key points to remember for interviews: explain tradeoffs, give Azure services mapping, and discuss performance implications.

## Chapter 5: Apache Spark Pools in Synapse

This chapter covers: Spark SQL vs DataFrame, Delta Lake, Optimizations, Streaming, ML workflows. It is written in a hybrid style — professional explanations with easy analogies and teaching tips.



### Section 1: Spark SQL vs DataFrame

Detailed explanation of Spark SQL vs DataFrame. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Spark SQL vs DataFrame lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 2: Delta Lake

Detailed explanation of Delta Lake. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Delta Lake lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 3: Optimizations

Detailed explanation of Optimizations. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Optimizations lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 4: Streaming

Detailed explanation of Streaming. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Streaming lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Section 5: ML workflows**

Detailed explanation of ML workflows. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

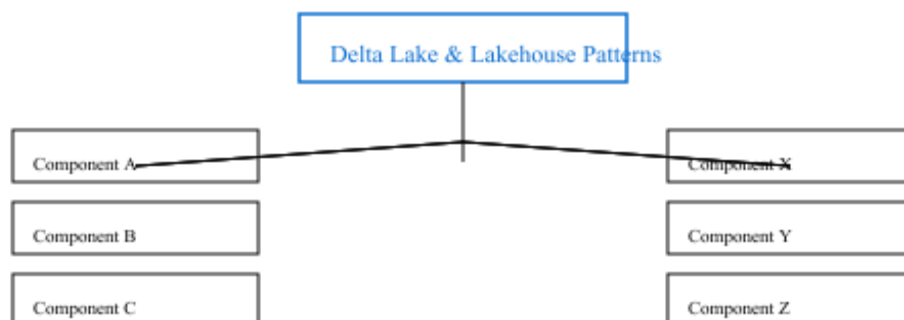
Lab: Perform the ML workflows lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Interview Prep & Quick Notes**

Key points to remember for interviews: explain tradeoffs, give Azure services mapping, and discuss performance implications.

## Chapter 6: Delta Lake & Lakehouse Patterns

This chapter covers: Delta log & ACID, Time travel, OPTIMIZE/ZORDER, Bronze/Silver/Gold, MERGE/CDC. It is written in a hybrid style — professional explanations with easy analogies and teaching tips.



### Section 1: Delta log & ACID

Detailed explanation of Delta log & ACID. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Delta log & ACID lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 2: Time travel

Detailed explanation of Time travel. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Time travel lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 3: OPTIMIZE/ZORDER

Detailed explanation of OPTIMIZE/ZORDER. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the OPTIMIZE/ZORDER lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 4: Bronze/Silver/Gold

Detailed explanation of Bronze/Silver/Gold. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Bronze/Silver/Gold lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Section 5: MERGE/CDC**

Detailed explanation of MERGE/CDC. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

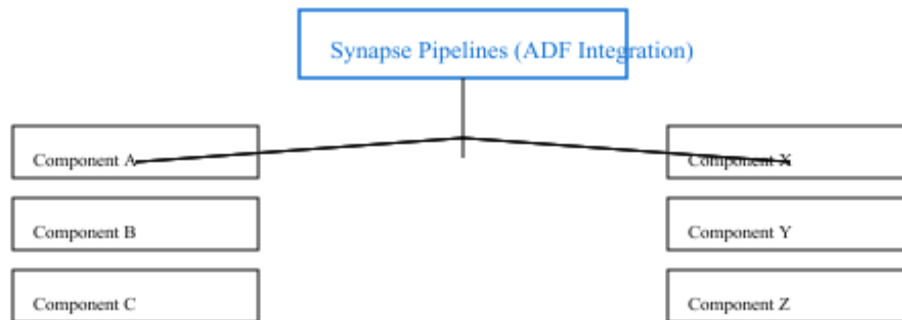
Lab: Perform the MERGE/CDC lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Interview Prep & Quick Notes**

Key points to remember for interviews: explain tradeoffs, give Azure services mapping, and discuss performance implications.

## Chapter 7: Synapse Pipelines (ADF Integration)

This chapter covers: Activities, Integration Runtimes, Mapping Data Flows, Triggers, CI/CD for pipelines. It is written in a hybrid style — professional explanations with easy analogies and teaching tips.



### Section 1: Activities

Detailed explanation of Activities. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Activities lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 2: Integration Runtimes

Detailed explanation of Integration Runtimes. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Integration Runtimes lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 3: Mapping Data Flows

Detailed explanation of Mapping Data Flows. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Mapping Data Flows lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 4: Triggers

Detailed explanation of Triggers. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Triggers lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Section 5: CI/CD for pipelines**

Detailed explanation of CI/CD for pipelines. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

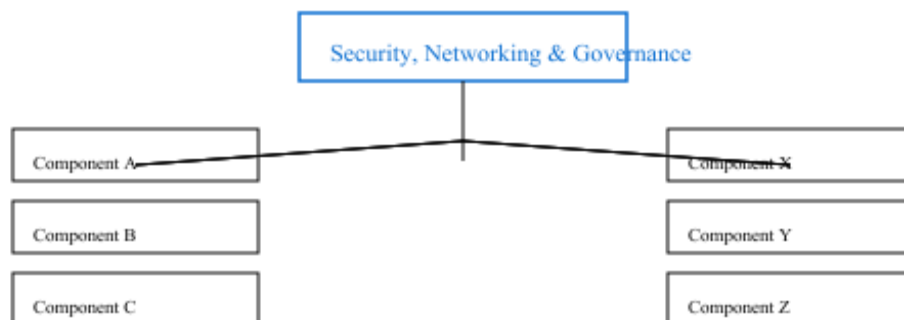
Lab: Perform the CI/CD for pipelines lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Interview Prep & Quick Notes**

Key points to remember for interviews: explain tradeoffs, give Azure services mapping, and discuss performance implications.

## Chapter 8: Security, Networking & Governance

This chapter covers: Authentication, RBAC, Private Endpoints, Encryption, Purview integration. It is written in a hybrid style — professional explanations with easy analogies and teaching tips.



### Section 1: Authentication

Detailed explanation of Authentication. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Authentication lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 2: RBAC

Detailed explanation of RBAC. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the RBAC lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 3: Private Endpoints

Detailed explanation of Private Endpoints. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Private Endpoints lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 4: Encryption

Detailed explanation of Encryption. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Encryption lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Section 5: Purview integration**

Detailed explanation of Purview integration. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

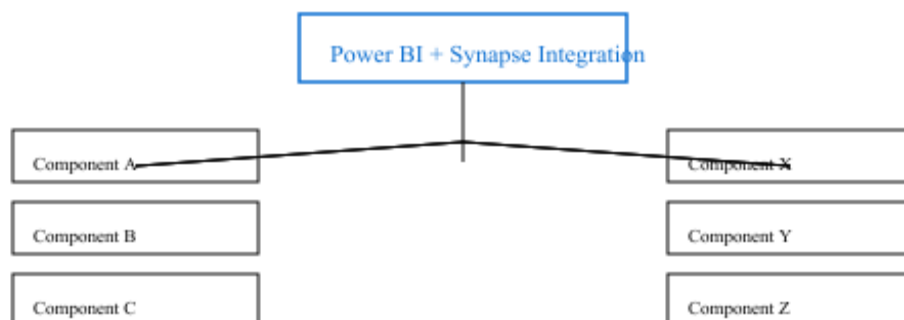
Lab: Perform the Purview integration lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Interview Prep & Quick Notes**

Key points to remember for interviews: explain tradeoffs, give Azure services mapping, and discuss performance implications.

## Chapter 9: Power BI + Synapse Integration

This chapter covers: DirectQuery vs Import, Performance tips, Using Serverless SQL, Materialized Views, Dashboard patterns. It is written in a hybrid style — professional explanations with easy analogies and teaching tips.



### Section 1: DirectQuery vs Import

Detailed explanation of DirectQuery vs Import. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

#### Code Example:

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https://path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

Lab: Perform the DirectQuery vs Import lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 2: Performance tips

Detailed explanation of Performance tips. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

#### Code Example:

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https://path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

Lab: Perform the Performance tips lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 3: Using Serverless SQL

Detailed explanation of Using Serverless SQL. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

**Code Example:**

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https:///path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

Lab: Perform the Using Serverless SQL lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## Section 4: Materialized Views

Detailed explanation of Materialized Views. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

**Code Example:**

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https:///path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

Lab: Perform the Materialized Views lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## Section 5: Dashboard patterns

Detailed explanation of Dashboard patterns. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

**Code Example:**

```
-- Sample SQL snippet SELECT TOP 10 * FROM OPENROWSET(BULK 'https:///path/*.parquet',  
FORMAT='PARQUET') AS rows;
```

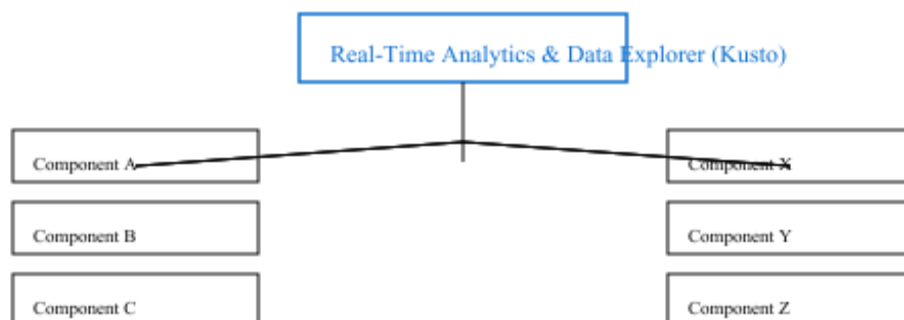
Lab: Perform the Dashboard patterns lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Interview Prep & Quick Notes**

Key points to remember for interviews: explain tradeoffs, give Azure services mapping, and discuss performance implications.

## Chapter 10: Real-Time Analytics & Data Explorer (Kusto)

This chapter covers: KQL basics, Data Explorer pool, Ingestion patterns, Time-series analysis, Alerting/monitoring. It is written in a hybrid style — professional explanations with easy analogies and teaching tips.



### Section 1: KQL basics

Detailed explanation of KQL basics. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the KQL basics lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 2: Data Explorer pool

Detailed explanation of Data Explorer pool. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Data Explorer pool lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 3: Ingestion patterns

Detailed explanation of Ingestion patterns. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Ingestion patterns lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 4: Time-series analysis

Detailed explanation of Time-series analysis. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Time-series analysis lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Section 5: Alerting/monitoring**

Detailed explanation of Alerting/monitoring. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

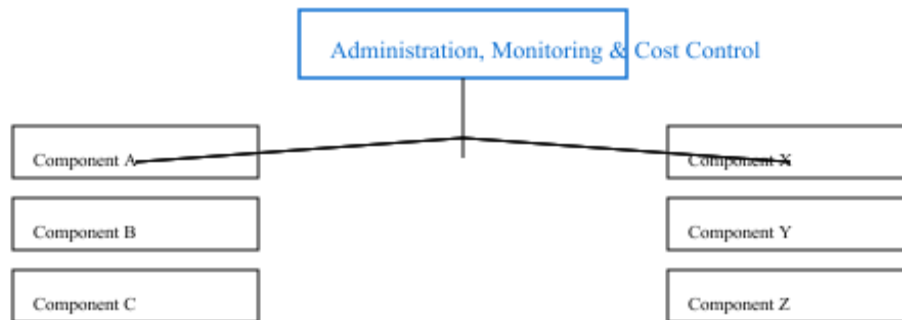
Lab: Perform the Alerting/monitoring lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Interview Prep & Quick Notes**

Key points to remember for interviews: explain tradeoffs, give Azure services mapping, and discuss performance implications.

## Chapter 11: Administration, Monitoring & Cost Control

This chapter covers: Workspace management, Monitoring jobs, Cost optimization, Auto-pause/scale, Alerting. It is written in a hybrid style — professional explanations with easy analogies and teaching tips.



### Section 1: Workspace management

Detailed explanation of Workspace management. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Workspace management lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 2: Monitoring jobs

Detailed explanation of Monitoring jobs. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Monitoring jobs lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 3: Cost optimization

Detailed explanation of Cost optimization. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Cost optimization lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 4: Auto-pause/scale

Detailed explanation of Auto-pause/scale. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Auto-pause/scale lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Section 5: Alerting**

Detailed explanation of Alerting. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

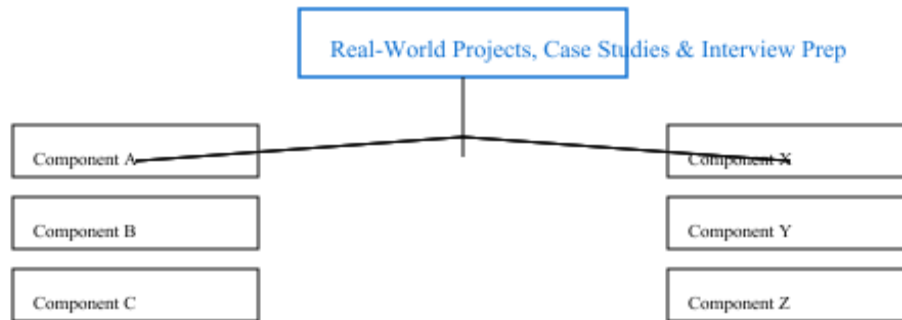
Lab: Perform the Alerting lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Interview Prep & Quick Notes**

Key points to remember for interviews: explain tradeoffs, give Azure services mapping, and discuss performance implications.

## Chapter 12: Real-World Projects, Case Studies & Interview Prep

This chapter covers: Design patterns, 10 project blueprints, Common pitfalls, Interview Q&A;, Capstone lab. It is written in a hybrid style — professional explanations with easy analogies and teaching tips.



### Section 1: Design patterns

Detailed explanation of Design patterns. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Design patterns lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 2: 10 project blueprints

Detailed explanation of 10 project blueprints. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the 10 project blueprints lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 3: Common pitfalls

Detailed explanation of Common pitfalls. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Common pitfalls lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

### Section 4: Interview Q&A;

Detailed explanation of Interview Q&A.; This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Interview Q&A; lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Section 5: Capstone lab**

Detailed explanation of Capstone lab. This section includes concepts, examples, best practices, and a mini-lab to reinforce learning.

Lab: Perform the Capstone lab lab. Steps: (1) Setup resources; (2) Execute sample; (3) Observe results; (4) Optimize.

## **Interview Prep & Quick Notes**

Key points to remember for interviews: explain tradeoffs, give Azure services mapping, and discuss performance implications.